



Resistor Value Classifier

AI-powered classification system that identifies resistor types and values using computer vision

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Initial Dataset Used For Prototype Training.

670

4-Band Resistors

Images with four color bands

600

5-Band Resistors

Images with five color bands

90

Classes

Unique resistor values

1,273

Total Images

Training and validation

Images sourced from Kaggle with varied lighting conditions and angles.

At least one addition large data set will be added

Three Model Approach

How the Pipeline Works

Instead of one model trying to do everything, we use three specialized models

1

Band Type Classifier

Looks at the image and decides: is this a 4-band or 5-band resistor?

2

Value Model

Based on that answer, the correct specialist model predicts the actual resistance value

This approach was taken because 4-band and 5-band resistors encode values differently, so one combined model would get confused.

All Three Models Share the Same Foundation

- All built on ResNet-18 using transfer learning — we reuse a network already trained on millions of images and fine-tune it for resistors
- Custom output layers added on top for our specific classes
- All take the same input: photos resized to 224×224

Architecture & Training

Model Architecture

- ResNet-18 convolutional neural network
- Transfer learning with fine-tuning
- Custom output layers for classification
- Dropout for regularization

Training Configuration

- Adam optimizer (adjusts the learning rate during training)
- Piecewise learning rate schedule
- GPU Accelerated Training
- Band-type classifier trains for 20 epochs since it only distinguishes 2 classes; value models train for 50 epochs because they must distinguish across many more resistor values

Input Processing

- Images resized to 224×224×3 (W x H x RGB)
- Automatic loading from folders
- Labels from folder / image names

Data Splitting

- 85% training set
- After training, the model classifies every image in the 15% validation set and compares predictions to true labels — accuracy is calculated as correct predictions divided by total images



Data & Augmentation

Images are organized in folders — the folder name tells the model what the resistor value is

Folders starting with 4B- are 4-band resistors, 5B- are 5-band

The code automatically strips out extra info from the files name, leaving just the value (e.g. 4B-10k-T5 becomes 10k)

85% of images used for training, 15% held back for testing accuracy

Augmentation — Why? In the real world, photos of resistors vary a lot — different angles, lighting, camera quality. Augmentation artificially creates that variety during training so the model doesn't just memorize the training images.

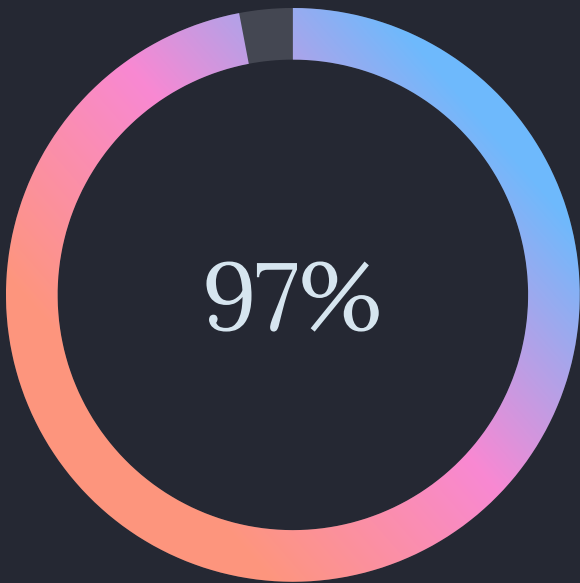
What we apply:

- *Position changes* — rotate, shift, zoom, flip
- *Lighting changes* — brightness and contrast adjustments, hue/color shifts
- *Quality changes* — blur, noise, simulate a low-res photo

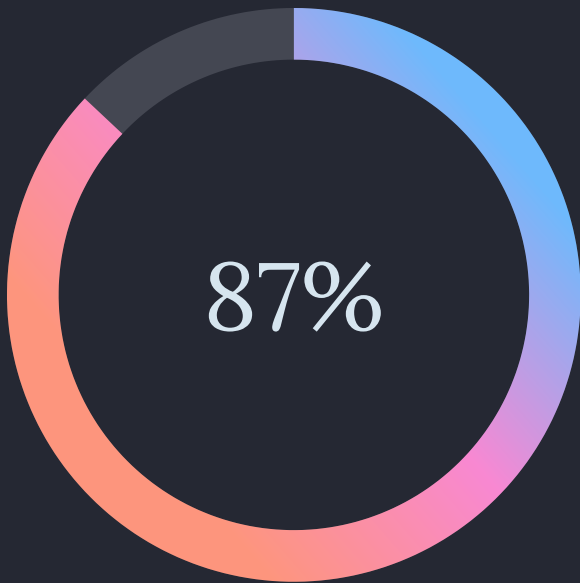
Training Results



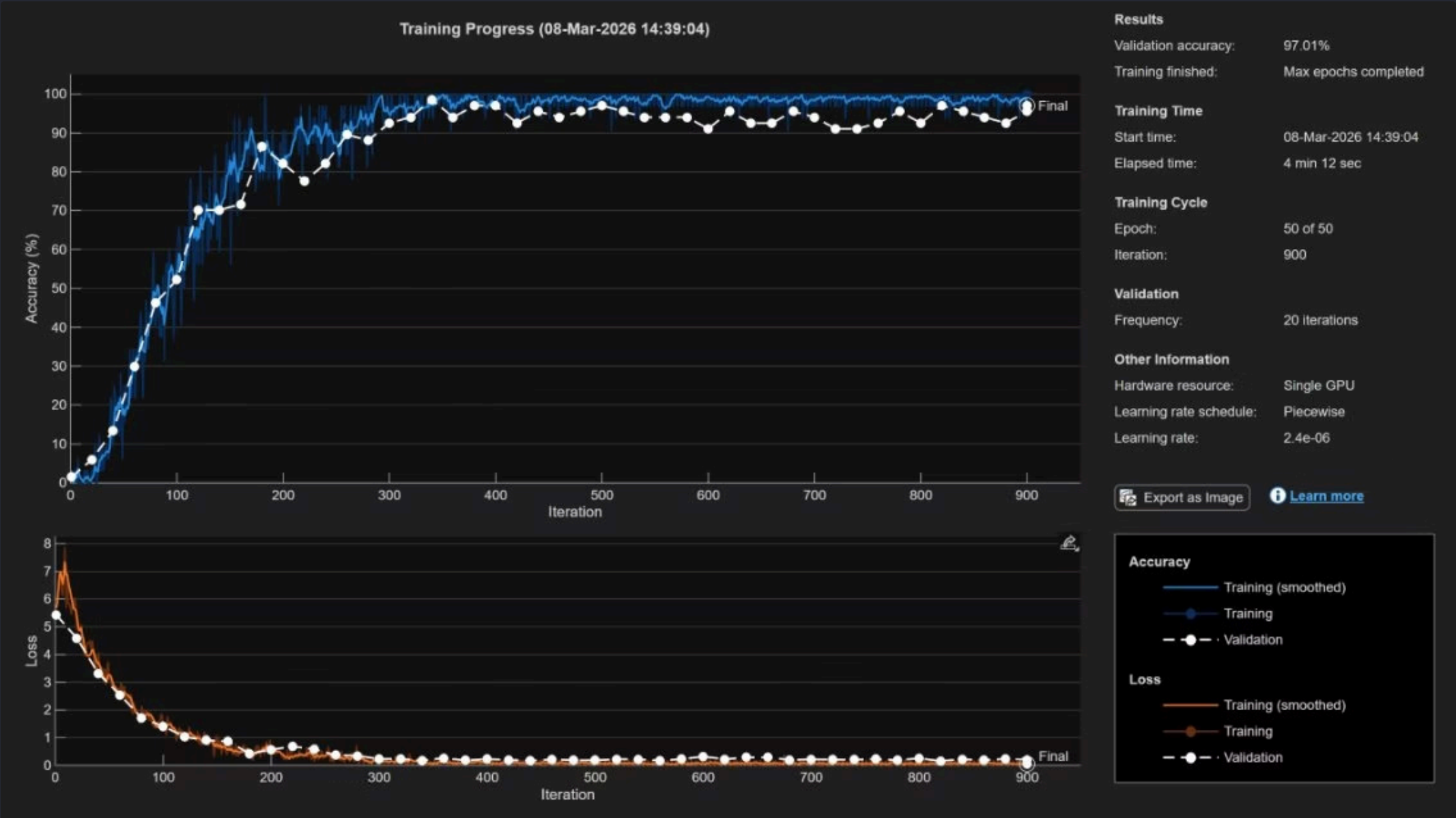
Validation Accuracy
Band Type Classifier



Validation Accuracy
4-Band Value Model



Validation Accuracy
5-Band Value Model



Confusion Matrix

98.82%

Overall Accuracy

Across 90 classes

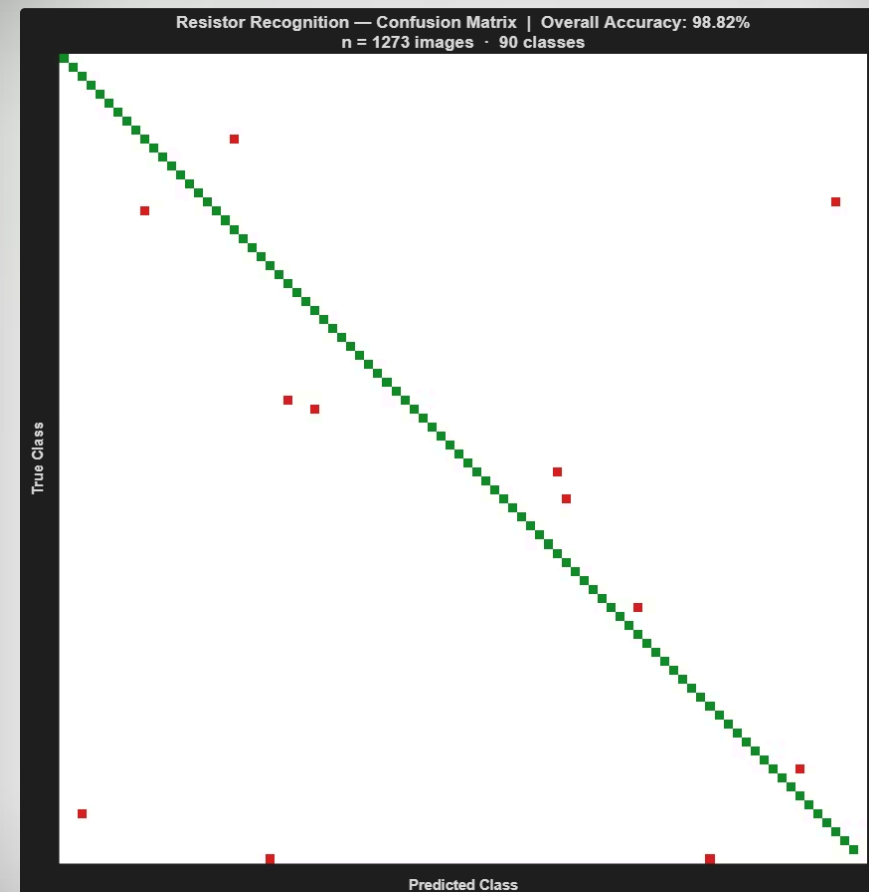
1,273

Test Images

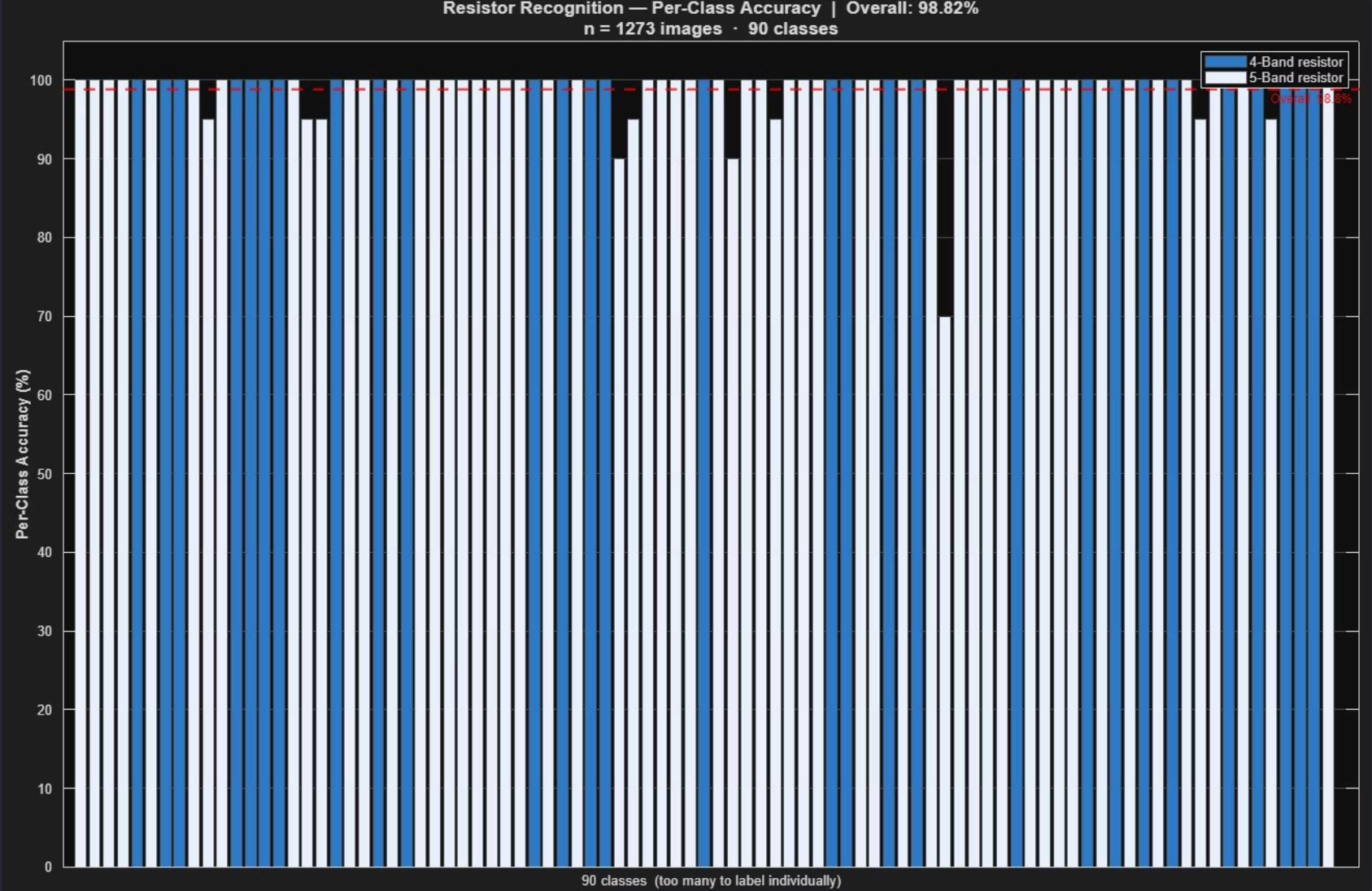
Used for evaluation

Most classes achieved 100% accuracy

With the rest have minimal misclassifications



Per-Class Accuracy



Running it

Input Image → Form Training Data Set

4-band resistor with Brown, red, orange, gold color bands (12k Ohms)

Analysis Output

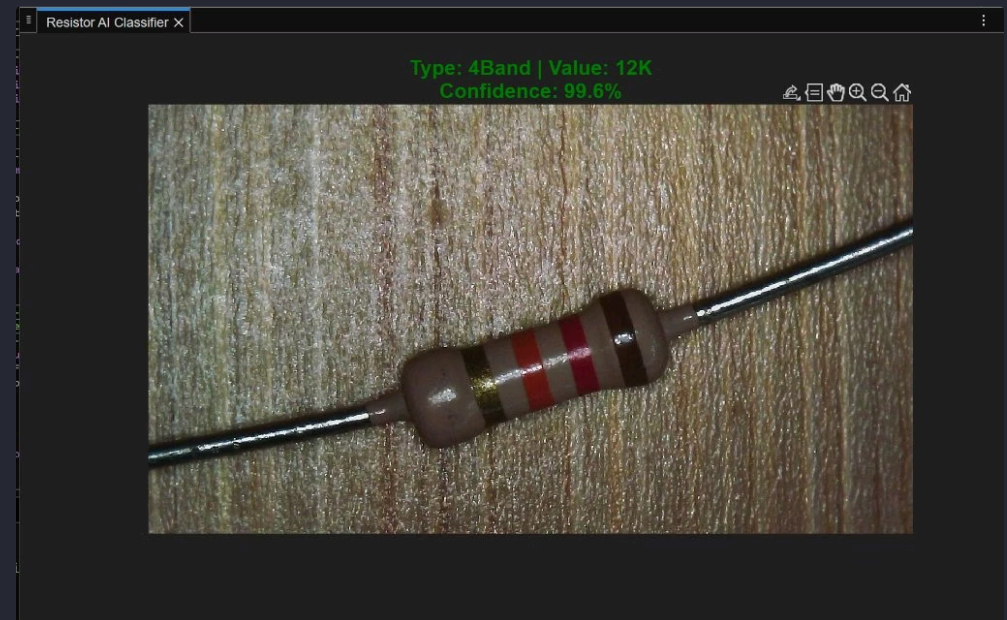
--- ANALYSIS START ---

Structure: 4Band (100% confidence)

Top Predictions:

1. 12K (99.6%)
2. 18K (0.3%)
3. 15K (0.0%)

MatLab Program



What's next?



Current Results

- Accuracy on our dataset: **98.82%**
- One class is **highly problematic** → needs deeper investigation
- **Performance significantly drops** on Images not from dataset



Planned Improvements

- **Expand dataset:** Add a second, larger, more diverse set of resistor images
- **Enhanced augmentation:** Apply additional image manipulations (lighting, rotation, noise, scale, blur) with **multiple passes per image**
- **Next model step:** Explore **other architectures** like a dedicated **CNN**



Goal

- Improve **robustness and real-world performance**
- Achieve **high accuracy on unseen resistor images**